discussion of these points, also because it would rely on a non-superficial mastering of physical arguments. Rather, we will follow the simpler logic of showing that the foundations of quantum mechanics can be deduced essentially by a single crucial fact, namely the *Heisenberg's realization of the uncertainty relations*, which affect the measurement of physical quantities at the microscopic level.

To fully appreciate the strength of Heisenberg intuition, we will start by a general revisitation of the mathematical structures and ideas underlying the foundations of classical mechanics.

1.2 Mathematical description of classical Hamiltonian systems

In order to realize the roots of the conflict between atomic physics and classical physics, we will isolate the basic structure underlying the mathematical description of a classical mechanical system.

Kinematics. The *configuration* or the *state* of a *classical Hamiltonian system* is (assumed to be) described by a set of canonical variables $\{q, p\}$, $q = (q_1, ..., q_n)$, $p = (p_1, ..., p_n)$, briefly by a point $P = \{q, p\} \in \Gamma \equiv$ phase space manifold. For simplicity, in the following, we will confine our discussion to the case in which Γ is compact. This is, e.g., the case in which the system is confined in a bounded region of space and the energy is bounded.

The physical quantities or *observables* of the system, clearly include the q's and p's and their polynomials and therefore, without loss of generality, we can consider as observables their sup-norm closure, i.e. (real) continuous functions $f(q, p) \in C_{\mathbf{R}}(\Gamma)$, (for a further extension see the remark after eq. (1.2.11)).

Every state P determines the values of the observables on that state and conversely, by the Stone-Weierstrass and Urysohn theorems, any $P \in \Gamma$ is uniquely determined by the values of all the observables on it (duality relation between states and observables).

Dynamics. The relation between the measurement of an observable f at an initial time t_0 and at any subsequent time t is given by the time evolution of the canonical variables

$$q \to q_t = q(t, q, p), \quad p \to p_t = p(t, q, p), \qquad f_t(q, p) \equiv f(q_t, p_t). \qquad (1.2.1)$$

The time evolution of the canonical variables is given by the Hamilton equations

$$\dot{q} = \frac{\partial H}{\partial p}, \; \dot{p} = -\frac{\partial H}{\partial q}, \qquad (1.2.2)$$

where $H = H(q,p)$ is the Hamiltonian. Under general conditions (typically if grad$H$ is Lipschitz continuous), for any initial data, the above system of equations has a unique solution local in time, which can be extended to all times under general conditions, e.g. if the surfaces of constant energy are compact [3].

The mathematical structures involved are the theory of functions (on phase space manifolds) and the theory of first order differential equations, defined by Hamiltonian flows on phase space manifolds.

From the above picture of elementary Hamiltonian mechanics one can extract the following algebraic structure.

I. **Algebraic properties of the classical observables.** The observable quantities associated to a classical system generate an abelian algebra $\mathcal{A}$ of real or more generally complex [4] continuous functions on the (compact) phase space (the product being given by the pointwise composition of functions $(fg)(x) = f(x)g(x)$ etc.). This algebra has an identity $\mathbf{1}$ given by the function $f = 1$ and a natural involution or $\star$ operation is defined by the ordinary complex conjugation, $f^*(x) = \bar{f}(x)$, so that $\mathcal{A}$ is a *$\star$-algebra with identity.* To each element $f \in \mathcal{A}$ one can assign a norm, $\| f \|$, given by the sup-norm

$$\| f \| = \sup_{x\in\Gamma} | f(x) |, \tag{1.2.3}$$

so that $\mathcal{A}$ is a Banach space with respect to this norm. The product is continuous with respect to the norm topology since

$$\| fg \| \leq \| f \| \, \| g \| \tag{1.2.4}$$

and therefore $\mathcal{A}$ is a *Banach $\star$-algebra.* Finally, the following property holds (*C^*-condition*)

$$\| f^* f \| = \| f \|^2. \tag{1.2.5}$$

Technically, an algebra with the above properties is called an *abelian C^*-algebra.*[5]

II. **States as linear functionals.** From an operational point of view, the identification of the states of a classical system with points of the phase space Γ relies on the unrealistic idealization, according to which the configuration of the system is sharply defined by measuring the canonical variables (typically positions and velocities) with infinite precision. Clearly, from a

[3] For a discussion of the existence theorems see V. Arnold, *Ordinary Differential Equations*, Springer 1992, and V. Arnold, *Mathematical Methods of Classical Mechanics*, Springer 1989.

[4] Such extension is both natural and convenient and in any case completely determined by the real subalgebra.

[5] For a beautiful account of the theory of abelian C^*-algebras see I.M. Gelfand, D.A. Raikov and G.E. Shilov, *Commutative Normed Rings*, Chelsea 1964; see also R.S. Doran and V.A. Belfi, *Characterization of C^*-Algebras*, Dekker 1986, esp. Chap. 2.

physical point of view it is more sensible to admit that in the preparation or detection of a state of a physical system a certain undeterminacy is unavoidable so that the configuration of the system at the initial time t_0 is known within a certain error, which inevitably propagates in time.

Since a state of the system is characterized by the measurements of the observables in that state, it is convenient to recall the operational meaning of such a procedure.

As it is well known, measurements with infinite precision are not possible and therefore the standard experimental way of associating a value of an observable f to a state ω is to perform replicated measurements of f, $m_1^{(\omega)}(f)$, $m_2^{(\omega)}(f)$, ..., $m_n^{(\omega)}(f)$, on the system in the given state ω or more generally on replicas of it and to compute the average

$$< f >_n^{(\omega)} \equiv [m_1^{(\omega)}(f) + m_2^{(\omega)}(f) + m_n^{(\omega)}(f)]/n.$$

The limit $n \to \infty$ (whose existence is part of the foundations of experimental physics) defines the *expectation of* f on the state ω

$$\omega(f) \equiv \lim_{n \to \infty} \; < f >_n^{(\omega)} \tag{1.2.6}$$

as *average of the results of measurements of* f in the state ω.

The coarseness affecting the measurements of f is given by

$$(\Delta_\omega f)^2 \equiv \omega((f - \omega(f))^2), \tag{1.2.7}$$

which indicates how much the results of measurements of f in the given state ω depart from the average $\omega(f)$; it is also called the *mean square deviation or variance* of f (relative to ω). More generally, all experimental information on the measurement of an observable f in the state ω are recorded in the expectations of the polynomials of f. [6]

This is the way the experimental results are recorded and the operational identification of a state of a physical system is therefore given by the set of expectation values of its observables. Since the expectation $\omega(f)$ of an observable f has the interpretation (and actually corresponds to the operational definition) of the average of the results of the measurements of f in the given state ω, it follows that such expectations are linear, i.e.

$$\omega(\lambda f_1 + \mu f_2) = \lambda \omega(f_1) + \mu \omega(f_2), \quad \forall f_1, f_2 \in \mathcal{A}, \quad \lambda, \mu \in \mathbf{C} \tag{1.2.8}$$

and satisfy the *positivity condition*, namely

$$\omega(f^* f) \geq 0, \quad \forall f \in \mathcal{A}. \tag{1.2.9}$$

[6] Such characterization of the measurements of f is somewhat related to the moment problem, for which a bound on the expectations of the polynomials of f is provided by the scale bound of the experimental apparatus associated to the measurements of f (such strict relations between observables and apparatuses yielding their measurements will be further discussed and exploited in the next section).

The positivity condition ($\omega((A+B)^*\,(A+B)) \geq 0$) implies the validity of Cauchy-Schwarz' inequality

$$|\omega(A^*B)| \leq \omega(A^*A)^{1/2}\,\omega(B^*B)^{1/2}, \quad \forall A, B \in \mathcal{A}, \tag{1.2.10}$$

and therefore $\omega(\mathbf{1}) > 0$ unless ω is the trivial state ($\omega = 0$ on $\mathcal{A}$). Thus, without loss of generality, given a (non-trivial) state ω one may always normalize it: $\omega \to \omega_{norm} = \omega(\mathbf{1})^{-1}\omega$, so that $\omega_{norm}(\mathbf{1}) = 1$.

Thus, in conclusion and quite generally, a classical system is defined by the abelian C^*-algebra $\mathcal{A}$ of its observables and a *state* of a classical system is a *normalized positive linear functional* ω on $\mathcal{A}$. A state ω on a C^*-algebra of continuous functions $C(X)$ on a compact (Hausdorff) space X is automatically continuous and therefore by the Riesz-Markov representation theorem [7] it defines a unique (regular Borel) measure μ_ω on X such that

$$\omega(f) = \int_X f\, d\mu_\omega, \quad \mu_\omega(X) = \omega(\mathbf{1}) = 1, \tag{1.2.11}$$

so that the expectations have a probabilistic interpretation [8]. Thus, the operational characterization of a state of a physical system leads to its description by a probability distribution rather than by a point of the phase space and the observables get the meaning of *random variables*.

The above considerations support the description of observables by continuous functions and justify the possible extension of the concept of observable to the pointwise limits of continuous functions, almost everywhere with respect to μ_ω.

Clearly, the above general concept of state contains as a very special case the definition of state in elementary mechanics; in fact, if μ_{P_0} is the probability measure concentrated on the point $P_0 = \{q_0, p_0\}$, i.e. for any measurable set S, $\mu_{P_0}(S) = 1$ if $P_0 \in S$ and $= 0$ otherwise, (namely μ_{P_0} is a Dirac δ function), then the corresponding state ω_{P_0} is

$$\omega_{P_0}(f) = \int_\Gamma f\, d\mu_{P_0} = f(P_0), \tag{1.2.12}$$

i.e. it is described by the point P_0. Such states are also called *pure states* since they cannot be written as convex linear combinations of other states (see also the next Sections). Clearly, for a pure state ω the variance vanishes, i.e. f takes a sharp value f_ω in the state ω, in the sense that all the measurements yield the same result, $f_\omega = \omega(f)$, i.e. there is no dispersion (such states are also called *dispersion free states*).

[7] A simple discussion is in M. Reed and B. Simon, *Methods of Modern Mathematical Physics*, Academic Press 1972, Vol. I (Functional Analysis), Chap. IV, Sect. 4.

[8] For an introduction to the theory of probability and in particular to the concept of random variable see, e.g. J. Lamperti, *Probability*, W.A.Benjamin 1966, esp. Chap. 1, and H.G. Tucker, *A Graduate Course in Probability*, Academic Press 1967, esp. Chaps. 1,2. A brief recollection is given in Sect. 2.4, below.

For the general states defined above the time evolution can be defined by duality in terms of the time evolution of the observables

$$\omega_t(f) \equiv \omega(f_t). \tag{1.2.13}$$

The above mathematical description of a state of a classical system is not only strongly suggested by operational arguments, concerning the measurement of physical quantities, but it is absolutely necessary for the mathematical and physical description of complex systems, i.e. when the number of degrees of freedom become very large, typically 10^{23} for thermodynamical systems. In this case, it is technically impossible to control an initial value problem for such a large number of variables and it is also physically unreasonable to measure all of them. Moreover, such idealistic description of a complex system is not what is required on physical grounds to account for the time evolution of physically realizable measurements. Thus, the mathematical and physical description of a complex system inevitably requires new mathematical ideas and structures with respect to those of classical analysis, namely the *theory of random variables.* This was indeed the revolutionary step taken by Boltzmann in laying the foundations of Classical Statistical Mechanics and in deriving classical thermodynamics from the mechanical properties of complex systems.

III. **Algebraic Dynamics**. Under general regularity conditions, in the concrete case of the canonical realization of a classical system, the time evolution $\{q, p\} \rightarrow \{q_t, p_t\}$ is continuous in time $t \in \mathbf{R}$, with a continuous dependence on the initial data at time t_0 and therefore it defines a one-parameter family of continuous invertible mappings $\alpha_{t_0,t}$ of $C(\Gamma)$ into itself,[9] which preserves all the algebraic relations, including the $*$-operation (by eq. (1.2.2) $\alpha_t(f\, g) = \alpha_t(f)\, \alpha_t(g)$, $\alpha_t(f^*) = (\alpha_t(f))^*$). A linear invertible mapping of a C^*-algebra into itself, which preserves the algebraic relations is called a $*$-automorphism (it follows from a general result that it necessarily preserves the norm, see e.g. Proposition 2.2.3 in the next chapter).

Quite generally, given an abelian C^*-algebra $\mathcal{A}$ of observables a time-translation invariant (reversible) dynamics, (i.e. one which depends only on the difference $t - t_0$), can be algebraically defined as a one-parameter group of $*$-automorphisms α_t of $\mathcal{A}$, $t \in \mathbf{R}$ and by duality one can define a one-parameter group of transformations α_t^* of states into states given by

$$\omega_t(A) \equiv (\alpha_t^* \omega)(A) \equiv \omega(\alpha_t(A)), \tag{1.2.14}$$

[9] In the case of non-regular dynamics, i.e. when the algebra of continuous functions on Γ is not stable under time evolution, one has to identify the observables with a larger C^*-algebra, than that of the continuous functions on Γ, in order to guarantee stability under time evolution (a necessary requirement for a reasonable physical interpretation).

(*time evolution of the states*). The abstract version of the continuity in time is that for any state ω, $\omega(\alpha_t(A))$, is continuous in time $\forall A \in \mathcal{A}$; technically α_t is said to be *weakly continuous.*

The recognition of the above mathematical structure at the basis of the standard description of classical systems suggests an abstract characterization of a classical (Hamiltonian) system, with no a priori reference to the explicit realization in terms of canonical variables, phase space, continuous functions on the phase space, etc. In this perspective, since a physical system is described in terms of measurements of its observables, one may take the point of view that a classical system is *defined* by its physical properties, i.e. by the algebraic structure of the set of its measurable quantities or observables, which generate an abstract abelian C^*-algebra $\mathcal{A}$ with identity. The states of the system being fully characterized by the expectations of the observables are described by normalized positive linear functionals on $\mathcal{A}$ and the dynamics is a one-parameter group of $*$-automorphisms of $\mathcal{A}$.

It is important to mention that quite generally, by the Gelfand-Naimark representation theorem [10], an (abstract) abelian C^*-algebra $\mathcal{A}$ (with identity) is isometrically isomorphic to the algebra of complex continuous functions $C(X)$ on a compact Hausdorff topological space X, where X is intrinsically defined as the Gelfand spectrum of $\mathcal{A}$.

From the point of view of general philosophy, the picture emerging from the Gelfand theory of abelian C^*-algebras has far reaching consequences and it leads to a rather drastic change of perspective. In the standard description of a physical system the geometry comes first: one first specifies the coordinate space, (more generally a manifold or a Hausdorff topological space), which yields the geometrical description of the system, and *then* one considers the abelian algebra of continuous functions on that space. By the Gelfand theory the relation can be completely reversed: one may start from the abstract abelian C^*-algebra, which in the physical applications may be the abstract characterization of the observables, in the sense that it encodes the relations between the physical quantities of the system, and then one reconstructs the Hausdorff space such that the given C^*-algebra can be seen as the C^*-algebra of continuous functions on it. In this perspective, one may say that the algebra comes first, the geometry comes later. The total equivalence between the two possible points of view indicates a purely algebraic approach to geometry: compact Hausdorff spaces are described by abelian C^*-algebras with identity, whereas if the algebra does not have an identity one has a locally compact Hausdorff space.

Non-commutative geometry is the structure emerging when the algebra is non-commutative. [11]

[10] For the convenience of the reader a brief outline of the Gelfand-Naimark theory is given in Appendix B.

[11] A. Connes, *Non Commutative Geometry*, Academic Press 1994.

1.3 General mathematical description of a physical system

In this section we argue that the structure of a C^*-algebra of observables and states is the suitable language for the mathematical description of a physical system in general (including the atomic systems), with no reference to classical mechanics and its standard paradigms.[12] For a more expanded and complementary discussion, also in relation with the Dirac-von Neumann principles of quantum mechanics, see Sect. 3.6, below.

I. **Observables.** From an operational point of view, a physical system consists of the set of states in which it can be prepared and it is defined by its physical properties, i.e. by the set $\mathcal{O}$ of the physical quantities (briefly called *observables*), which can be measured on its states, and by the relations between them. Each observable has to be understood as characterized by a concrete physical apparatus yielding its measurements.[13]

For any $A \in \mathcal{O}$ and $\lambda \in \mathbf{R}$, one can define the observable λA as the observable measured by rescaling the apparatus (e.g. the pointer scale) by λ. By similar considerations one justifies the existence of elementary functions of an observable like the powers (with the standard elementary properties): if $A \in \mathcal{O}$, A^2 may be interpreted as the observable associated with squaring the apparatus scale (equivalently, by squaring each result of measurement given, e.g., by each pointer reading). Similarly, one defines the powers A^m and their products $A^m A^n \equiv A^{m+n}$. It follows from this definition that A^0 is the observable whose results of measurements always take the value 1, independently of the state on which the measurement is done. In the same way, one defines a polynomial of A as the observable obtained by taking as the new apparatus scale the given polynomial function of the scale for A.

An element $A \in \mathcal{O}$ is said to be *positive* if all the *results of measurements* of A are positive numbers. By the operational definition of elementary functions of A, this implies that (and it is actually equivalent to) A is of the form $A = B^2$, $B \in \mathcal{O}$. Clearly, given an observable A and two positive polynomials, $\mathcal{P}_1(A)$, $\mathcal{P}_2(A)$, their product is also positive.

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[12]The C^*-algebraic approach to classical and quantum physics has been pioneered by I. Segal, Ann. Math.(2) **48**, 930 (1947); *Mathematical Problems of Relativistic Physics*, Am. Math. Soc. 1963; see also P. Jordan, J. von Neumann and E.P. Wigner, Ann. Math. **35**, 29 (1934) for an early proposal and G.G. Emch, *Algebraic Methods in Statistical Mechanics and Quantum Field Theory*, Wiley-Interscience 1972, esp. Chap. 2, for a historical review and a systematic treatment.

[13]The possibility that two distinct experimental apparatuses effectively define the same observable will be discussed below. It is convenient to deal with dimensionless observables, whose measurements are defined as ratios with respect to a set of reference measurements (e.g. of length, mass etc.); such a choice of scale is implicit in each physical apparatus.

II. **States**. A *state* ω *of a physical system* defines, $\forall A \in \mathcal{O}$, the corresponding *expectation* $\omega(A)$; it is operationally obtained by performing replicated measurements of A when the system is in the state ω, or (better) in identically prepared states, and by taking the *average over the results of measurements.* The existence of such an average, in the limit of large number of measurements, is at the basis of experimental physics. Thus, ω defines a *real functional* on $\mathcal{O}$.

By the operational definition of $\omega(A)$ it easily follows that ω is a homogeneous functional

$$\omega(\lambda A) = \lambda\,\omega(A), \quad \forall \lambda \in \mathbf{R}, \tag{1.3.1}$$

and that $\omega(A^n + A^m) = \omega(A^n) + \omega(A^m)$.

The realization that the operational way of characterizing a state is in terms of its expectations of the observables implies that a state ω is completely characterized by the set of its expectations $\omega(A)$, when A varies over $\mathcal{O}$, and therefore two states yielding the same expectations must be identified, i.e.

$$\omega_1(A) = \omega_2(A), \quad \forall A \in \mathcal{O}, \tag{1.3.2}$$

must imply $\omega_1 = \omega_2$, (briefly the observables separate the states).

On the other hand, if we put at the basis of the mathematical description of a physical system the fact that the experimental way of identifying an observable is in terms of its expectations on the states, then two observables A and B having the same expectations on all the states, $\omega(A) = \omega(B)$, $\forall\omega$, cannot be distinguished. Such a property of the states, of completely characterizing the observables and their relations, can be viewed as a *completeness of the states* with respect to the observables. This means that the states define an equivalence relation, denoted by $\sim$, between the elements of $\mathcal{O}$: $A \sim B$, if $\omega(A) = \omega(B)$ for all the physical states ω, (for example two distinct experimental apparatuses may effectively define the same observable). Two equivalent observables must therefore be identified and in the following the set $\mathcal{O}$ of observables will always denote the corresponding set of equivalence classes.

The definition of the zero-th power of an observable A implies that for any physical state $\omega(A^0) = 1$, and therefore all the zero-th powers of observables fall in the same equivalence class, which will be called the identity and denoted by $\mathbf{1}$. Clearly, for any state

$$\omega(\mathbf{1}) = 1, \tag{1.3.3}$$

i.e. a state of a physical system is a *normalized real functional* on $\mathcal{O}$. Furthermore, the property $A^n A^m = A^{m+n}$ gives

$$A\,\mathbf{1} = A\,A^0 = A = \mathbf{1}A.$$

As we have also seen in the previous section, a crucial *positivity* property must be satisfied by the states. Since the expectation $\omega(A)$ is the *average* over the results of measurements in the state ω, it follows that, if A is positive, then, for any state ω,

$$\omega(A) = \omega(B^2) \geq 0, \tag{1.3.4}$$

i.e. any state ω is a *normalized positive functional* on $\mathcal{O}$.

By the completeness property of the states in identifying the observables, all the properties of an observable A have to be encoded in its expectation values and therefore, in particular, the positivity of an observable A has to be equivalently described by the positivity of all its expectations, $\omega(A) \geq 0$ for all ω. In particular, a polynomial $\mathcal{P}(A)$ of an observable A is *positive* if $\omega(\mathcal{P}(A)) \geq 0$, for all physical states ω.

III. C^***-algebraic structure**. As discussed above, an observable A is defined in terms of a concrete experimental apparatus, which yields the numerical results of measurements in any state and each concrete experimental apparatus has inevitable limitations implying a scale bound (e.g. the set of positions which the pointer can take is bounded) independent of the state on which the measurement is performed. Then, the results of measurements of A in the various states is a bounded set of numbers, with the bound being related to the scale bound of the associated experimental apparatus.

To each observable A it is then natural to associate the finite positive number

$$||A|| \equiv \sup_{\omega} |\omega(A)| < \infty. \tag{1.3.5}$$

Clearly, by the homogeneity of the states

$$||\lambda A|| = |\lambda| \, ||A||, \quad \forall \lambda \in \mathbf{R}. \tag{1.3.6}$$

Moreover, since the states separate the observables, $||A|| = 0$ implies $A = 0$.

From the definition of $||A||$ and positivity it follows that

$$||A^2|| = ||A||^2. \tag{1.3.7}$$

In fact, by definition, for any physical state ω, $\omega(||A||\mathbf{1} \pm A) \geq 0$, so that $||A||\mathbf{1} \pm A$ are both positive; then $(||A||\mathbf{1} - A)(||A||\mathbf{1} + A)$ is a positive polynomial of A and

$$||A||^2 - \omega(A^2) = \omega((||A||\mathbf{1} - A)(||A||\mathbf{1} + A)) \geq 0, \quad \forall \omega,$$

which implies $||A||^2 \geq ||A^2||$.

On the other hand, the positivity of

$$(||A||\mathbf{1} \pm A)^2 = ||A||^2 + A^2 \pm 2\,||A||A$$

implies that for any state ω

$$2\,||A||\;|\omega\,(A)| \;\leq\; ||A||^2 + \omega\,(A^2) \leq ||A||^2 + ||A^2||$$

and therefore $||A||^2 \;\leq\; ||A^2||$.

The duality relation between observables and states allows to display and define linear structures in $\mathcal{O}$. We have already argued that the sum of polynomials of one observable has a well defined operational meaning; actually for a larger class of pairs A, B, (e.g. the kinetic and the potential energy) the sum can be defined in the sense that there is an observable $C \in \mathcal{O}$ such that

$$\omega\,(C) = \omega(A)\; + \omega\,(B) \equiv \omega\,(A + B), \quad \forall\omega \tag{1.3.8}$$

and (by the duality relation between states and observables) one may write $C = A + B \in \mathcal{O}$.

For arbitrary pairs, A, B, the sum defined by the expectations (1.3.8) may not correspond to an element of $\mathcal{O}$ and therefore its introduction leads to an extension of $\mathcal{O}$, on which the states are positive linear functionals. Moreover, the definition of the powers $(A + B)^n$ may not have a direct operational meaning; then, the possibility, adopted in the sequel, of introducing them for *any* pair A, B, with the same algebraic properties of the powers of the generating observables and the extension of the states to them as *positive linear functionals* is a non-trivial extrapolation over the strict physically motivated structure. The so-obtained extension of $\mathcal{O}$ will still be denoted by $\mathcal{O}$.

Clearly, from eqs. (1.3.5), (1.3.8), $||A + B||$ is well defined and

$$||A + B|| \leq\; ||A||\; + \;||B||. \tag{1.3.9}$$

Thus $||\;||$ is a norm on $\mathcal{O}$, which becomes a pre-Banach space.

By the definition of the norm and positivity it follows that

$$\omega(A^2) \leq \omega(A^2) + \omega(B^2) = \omega(A^2 + B^2) \leq ||A^2 + B^2||,$$

so that

$$||A^2|| \leq ||A^2 + B^2||. \tag{1.3.10}$$

For technical reasons, it is convenient to consider the norm completion of $\mathcal{O}$, so that by a standard procedure one gets a real Banach space.

By definition of the norm, any state of the physical system satisfies

$$|\omega(A)| \leq ||A||, \tag{1.3.11}$$

i.e. any state is continuous with respect to the topology defined by the norm (*norm topology*) and therefore it has a unique continuous extension to the norm completion of $\mathcal{O}$, hereafter still denoted by $\mathcal{O}$.

The powers of the sum $A + B$ allow to define the following *symmetric product*, also called Jordan product

$$A \circ B \equiv \frac{1}{2}((A+B)^2 - A^2 - B^2) = B \circ A, \tag{1.3.12}$$

which however is not guaranteed to be distributive and associative.

It is easy to check that for the elements of the polynomial algebra generated by an observable A the symmetric product coincides with the algebraic product; in particular $A \circ A^n = A^{n+1}$.

The so-obtained structure is close to that advocated by Jordan, the so-called *Jordan Algebra*, [14] for which no topological structure is assumed, but the symmetric product is assumed to be distributive and weakly associative, i.e.

$$A^2 \circ (B \circ A) = (A^2 \circ B) \circ A.$$

Actually, distributivity follows from the (mild looking) assumption that the symmetric product is homogeneous, i.e.

$$A \circ (\lambda B) = \lambda(A \circ B), \quad \lambda \in \mathbf{R} \tag{1.3.13}$$

and then, by symmetry, also $(\lambda A) \circ B = \lambda(A \circ B)$.

Such a property is certainly satisfied when A and B are polynomial functions of the same observable C, since then a (distributive and associative) product is defined and $A \circ B = \frac{1}{2}(AB + BA)$; the extension to the general case looks like a reasonable assumption.[15]

[14] P. Jordan, Zeit f. Phys. **80**, 285 (1933); P. Jordan, J. von Neumann and E.P. Wigner, Ann. Math. **35**, 29 (1934); L.J. Page, Jordan Algebras, in *Studies in Modern Algebra*, A.A. Albert ed., Prentice Hall 1963; N. Jacobson, *Structure and Representations of Jordan Algebras*, Am. Math. Soc. 1968; H. Upmeier, *Jordan Algebras in Analysis, Operator theory and Quantum Mechanics*, Am. Math. Soc. 1980; H. Hanche-Olsen and E. Størmer, *Jordan Operator Algebras*, Pitman 1984.
A Jordan algebra is said to be special if the symmetric product arises from an associative product: $A \circ B = \frac{1}{2}(AB + BA)$; otherwise it is said to be exceptional. Actually, there is only one simple exceptional Jordan algebra and it has dimension 27, so that it does not appear suitable for the description of quantum systems (for an updated account, see K.McCrimmon, *A Taste of Jordan Algebras*, Springer 2004). For the role of Jordan algebras in the formulation of quantum mechanics see the general review by A.S. Wightman, Hilbert Sixth Problem: Mathematical Treatment of the Axioms of Physics, in *Proceedings of Symposia in Pure Mathematics*, Vol. 28, Am. Math. Soc. 1976).

[15] G.G. Emch, *Algebraic Methods in Statistical Mechanics and Quantum Field Theory*, Wiley-Interscience 1972, pp. 44-47.

A simple calculation shows that homogeneity implies distributivity. In fact, eq. (1.3.12) gives

$$(A+B)^2 = A^2 + B^2 + 2A \circ B, \tag{1.3.14}$$

$$(A-B)^2 = A^2 + B^2 + 2A \circ (-B) = A^2 + B^2 - 2A \circ B$$

and therefore $A \circ B = \frac{1}{4}((A+B)^2 - (A-B)^2)$, and

$$A^2 + B^2 = \frac{1}{2}((A+B)^2 + (A-B)^2). \tag{1.3.15}$$

Then, by eq. (1.3.14)

$$2(A+B) \circ C - 2A \circ C - 2B \circ C$$

$$= [(A+B+C)^2 + A^2] + [B^2 + C^2] - [(A+B)^2 + (A+C)^2] - (B+C)^2,$$

and eq. (1.3.15) applied to the three sums of squares in square brackets gives the vanishing of the r.h.s.

Furthermore, distributivity of the symmetric product and positivity of the states imply

$$0 \leq \omega\,((A+\lambda B)^2) = \omega\,((A+\lambda B) \circ (A+\lambda B))$$
$$= \omega\,(A^2) + \lambda^2 \omega\,(B^2) + 2\lambda\omega\,(A \circ B), \quad \forall \lambda \in \mathbf{R},$$

so that

$$|\omega\,(A \circ B)| \leq \omega\,(A^2)^{1/2}\,\omega\,(B^2)^{1/2} \tag{1.3.16}$$

and

$$||A \circ B|| \leq ||A||\;||B||. \tag{1.3.17}$$

As shown by Jordan, von Neumann and Wigner, weak associativity follows from $A^{n+1} = A \circ A^n$ and from the property of *formal reality*, which reads $\sum_i A_i^2 = 0 \;\Rightarrow\; A_i = 0$ and is implied by an obvious extension of eq. (1.3.10). Thus, the so-obtained structure is a Jordan algebra, actually a *Jordan Banach (JB) algebra*, namely a Jordan algebra which is also a Banach space with a norm satisfying eqs. (1.3.7), (1.3.10) and (1.3.17).

The structure discussed above is also rather close to that advocated by Segal for the description of quantum systems (*Segal system*), [16] for which distributivity of the symmetric product is not assumed, but the following continuity properties of the powers are required:

i) The square is continuous in the norm topology, i.e. $A_n \to A$ implies $A_n^2 \to A^2$,

ii) $||A^2 - B^2|| \leq \max\,(||A||^2, ||B||^2)$.

[16] I. Segal, Ann. Math.(2), **48**, 930 (1947).

The continuity properties i) assumed by Segal follow easily from distributivity and from eq. (1.3.17); in fact, by distributivity one has $A_n^2 - A^2 = (A_n + A) \circ (A_n - A)$ and therefore

$$||A_n^2 - A^2|| \leq ||A_n - A|| \, (||A_n - A|| + ||2A||),$$

which implies i).

Equation ii) is derived by using the positivity of the squares and of the states, the algebraic relation $|a^2 - b^2| \leq \max\,(a^2, b^2)$, $\forall a, b \in \mathbf{R}$, the definition of the norm, eq. (1.3.5), and eq. (1.3.7):

$$|\omega(A^2) - \omega(B^2)| \leq \max\,(\omega(A^2),\, \omega(B^2)) \leq \max\,(||A^2||,\, ||B^2||)$$
$$= \max\,(||A||^2,\, ||B||^2).$$

As shown by Segal, the above structure allows to recover most of the mathematical basis for the description of quantum systems, like the concept of compatible observables, the joint probability distribution for compatible observables etc. (see the above quoted book by Emch for an extensive discussion).

However, the mathematical language becomes easier if the so-obtained (distributive) Segal system, $\mathcal{O}$, which is also a JB algebra, is embedded in a C^*-algebra $\mathcal{A}$. This means that the elements of $\mathcal{O}$ generate an associative algebra $\mathcal{A}$ such that
1) the symmetric product arises from the associative (but not necessarily commutative) product in $\mathcal{A}$, i.e. $\forall A, B \in \mathcal{O}$

$$A \circ B = \frac{1}{2}\,(AB + BA),$$

2) an involution is defined on $\mathcal{A}$ with the properties that $\forall A$, $B \in \mathcal{O}$,

$$(\lambda A + \mu B)^* = \bar{\lambda} A + \bar{\mu} B, \qquad \forall \lambda, \mu \in \mathbf{C},$$

$$(AB)^* = BA,$$

3) $\forall A \in \mathcal{A}$, A^*A is positive and the states can be extended from $\mathcal{O}$ to $\mathcal{A}$ by linearity as linear functionals, with the natural extension of positivity

$$\omega\,(A^*A) \geq 0, \quad \forall A \in \mathcal{A},$$

and with the properties

$$||AB|| = \sup_\omega |\omega(AB)| \leq ||A||\,||B||, \quad ||A^*A|| = ||A^*||\,||A||. \tag{1.3.18}$$

Positivity, $\omega\,((\lambda A + \mathbf{1}\,)^*(\lambda A + \mathbf{1}\,)) \geq 0$, implies

$$\omega\,(A^*) = \overline{\omega(A)}\,, \qquad ||A^*|| = ||A||, \quad \forall A \in \mathcal{A}. \tag{1.3.19}$$

The so-obtained extension $\mathcal{A}$has the properties of a C^*-algebra with identity $\mathbf{1}$, $\mathcal{O}$ is the subset of $*$-invariant (also called *self-adjoint*) elements (briefly $\mathcal{O}$ is a Jordan subalgebra of $\mathcal{A}$) and $\mathcal{A}$ is generated by $\mathcal{O}$.[17] A JB algebra which is isometrically isomorphic to a Jordan subalgebra of a C^*-algebra is called a JC algebra.

A Segal system or a JB algebra which allows such an embedding is called special and exceptional otherwise.[18] In order to discuss the possibility of such an embedding, we start by defining the *complexification* $\mathcal{O}^{\mathbf{C}}$ of $\mathcal{O}$:

$$\mathcal{O}^{\mathbf{C}} = \{A + iB;\ A, B \in \mathcal{O}\}.$$

The symmetric product is easily extended by distributivity

$$(A + iB) \circ (C + iD) = (A \circ C - B \circ D) + i\,(A \circ D + B \circ C),$$

and the states extend by linearity: $\omega(A + iB) = \omega(A) + i\,\omega(B)$. The so-extended states have the following properties

$$\omega(A + i\,B) = \overline{\omega(A - i\,B)}, \quad \omega((A + iB) \circ (A - iB)) = \omega(A^2 + B^2) \geq 0. \tag{1.3.20}$$

An involution is defined on $\mathcal{O}^{\mathbf{C}}$ by $(A + iB)^* \equiv A - i\,B$ and one has

$$||A + iB|| \equiv \sup_\omega |\omega(A + iB)| = ||A - iB||, \quad \text{i.e.} \quad ||C|| = ||C^*||, \quad \forall C \in \mathcal{O}^{\mathbf{C}}.$$

The following theorem classifies the exceptional JB algebras.[19]

Theorem 1.3.1 *Any JB algebra $\mathcal{O}$ contains a unique exceptional ideal J such that $\mathcal{O}/J$ is a JC algebra and J is characterized by the property that every factor representation of J is onto the 27-dimensional Albert algebra.*

Since the Albert algebra appears too poor, and therefore not suitable, for the mathematical description of the observables of a physical system, in the following we shall assume that the observables define a special distributive Segal system, equivalently a JC algebra, and, therefore generate a C^*-algebra.

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[17]For an introduction to C^*-algebras see e.g. M. Takesaki, *Theory of Operator Algebras*, Vol. I, Springer 1979, Chap. I; R.V. Kadison and J.R. Ringrose, *Fundamentals of the Theory of Operator Algebras*, Vol. I, Academic Press 1983, Chap. 4; O. Bratteli and D.W. Robinson, *Operator Algebras and Quantum Statistical Mechanics*, Vol. I, Springer 1987, Sects. 2.1- 2.3; the basic textbook is J. Dixmier, *C^*-algebras*, North-Holland 1977. For the convenience of the reader a few basic notions about C^*-algebras are presented in the Appendices.

[18]Necessary and sufficient conditions for the existence of such an extension, have been given by D. Lowdenslager, Proc. Amer. Math. Soc. **8**, 88 (1957). For physically motivated conditions see E.M. Alfsen and F.W. Schultz, *Geometry of State Spaces of Operator Algebras*, Birkhäuser 2003, but their physical interpretation is not transparent.

[19]H. Hanche-Olsen and E. Størmer, *Jordan Operator Algebras*, Pitman 1984, Chap. 7, Theor. 7.2.3.

The arguments discussed in this section do not pretend to prove as a mathematical theorem that the general physical requirements on the set of observables necessarily lead to a C^*-algebraic structure, (the existence of the sums for all pairs $A, B \in \mathcal{O}$ and the homogeneity of the Jordan product lying beyond the strict operational setting), but they should provide sufficient motivations in favor of it (for a more detailed discussion see Sect. 3.6 below).

In any case, the above mathematical structure is by far more general than the concrete structure discussed in Sect. 2 for classical systems. Furthermore, the property that the observables generate a C^*-algebra is also implied by the much stronger Dirac-von Neumann axiom that the observables are described by the hermitian operators in the Hilbert space of states. A more detailed discussion of the relation with the Dirac-von Neumann axioms is given in Sect. 3.6.

As we shall see, the full mathematical setting of quantum mechanics can be derived with a very strict logic solely from the C^*-algebraic structure of the observables and the *operational information of non-commutativity codified by the Heisenberg uncertainty relations* (Sect. 2.1). In this way one has a (in our opinion better motivated) alternative to the Dirac-von Neumann axiomatic setting, which can actually be derived through the GNS theorem 2.2.4, the Gelfand-Naimark theorem 2.3.1 and von Neumann theorem 3.2.2 (see the discussion of the principles of quantum mechanics in Sect. 3.6). For these reasons we adopt the following mathematical framework:

1. A physical system is *defined* by its *C^*-algebra $\mathcal{A}$ of observables* (with identity).

2. The states of the given physical system are identified by the measurements of the observables, i.e. a *state* is a *normalized positive linear functional on $\mathcal{A}$*. The set $\mathcal{S}$ of physical states separates the observables, technically one says that $\mathcal{S}$ is *full*, and conversely the observables separate the states.

In the mathematical literature, given a C^*-algebra $\mathcal{A}$, any normalized positive linear functional on it is by definition a state; here we allow the possibility that the set $\mathcal{S}$ of states with physical interpretation (briefly called physical states) is full but smaller than the set of all the normalized positive linear functionals on $\mathcal{A}$.

Quite generally, given an abstract C^*-algebra $\mathcal{A}$, with identity $\mathbf{1}$, a positive linear functional ω on $\mathcal{A}$ is necessarily continuous with respect to the topology of the preassigned norm which makes $\mathcal{A}$ a C^*-algebra (see Appendix C, Proposition 1.6.3):

$$|\omega(A)| \leq ||A|| \, \omega(\mathbf{1}).$$

Also, $\omega(A) \geq 0, \forall\omega$ implies $A = B^*B$ (Proposition 1.6.2). In the above presentation, these properties were obtained on the basis of the operational definition of states and observables.

The spectrum $\sigma(A)$ of an element $A \in \mathcal{A}$ is the set of all λ such that $\lambda\mathbf{1} - A$ does not have a two-sided inverse in $\mathcal{A}$. This is the purely algebraic version of the standard definition of spectrum for operators in a Hilbert space. An element A is said to be *normal* if it commutes with its adjoint A^*. If A is normal and $\lambda \in \sigma(A)$, then there exists at least one positive linear functional ω such that $\omega(A) = \lambda$ (see Appendix C). Thus the spectrum of a normal element A is a set of possible expectations of A. This implies that the set of all positive linear functionals on $\mathcal{A}$ separate the normal elements of $\mathcal{A}$ and therefore all the elements of $\mathcal{A}$, since any A can be written as a complex linear combination of normal elements $A = (A + A^*)/2 - i(iA - iA^*)/2$.